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Head CT Scan Motion Artifact Correction via Diffusion-Based Generative Models

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Abstract

Head motion is a major source of image artifacts in head computed tomography (CT), degrading the image quality and impacting diagnosis. Image-domain-based motion correction is practical for routine use since it doesn't rely on hard-to-obtain CT projection data. However, existing convolutional neural network (CNN)-based methods tend to over-smooth images, particularly in cases of moderate to severe 3D motion artifacts. Motivated by the improved image quality and more stable training of diffusion-based generative models, we propose a novel 3D head CT motion correction approach based on conditional diffusion, named HeadMotion-EDM(HM-EDM). This approach has three features. Firstly, we utilize motion-corrupted images as the conditional input. Secondly, we leverage the advanced Elucidated Diffusion Model (EDM) framework, which integrates several pivotal engineering improvements in the diffusion model and significantly expedites the sampling process. Thirdly, we design an efficient 3D-patch-wise training method for 3D CT data. Comparative studies demonstrate that our approach surpasses CNN-based techniques as well as the denoising diffusion probabilistic model (DDPM) in both simulation and phantom studies. Furthermore, radiologists reviewed the results of applying HM-EDM to real-world portable head CT scans, showing its effectiveness in eliminating motion artifacts and improving diagnostic value.

Keywords

Head CT; Motion correction; Diffusion Model

1 Introduction

Patient head motion, especially in pediatric patients or those with head trauma or stroke, is a common source of image artifacts in computed tomography (CT) of the head[1]. Motion causes data inconsistency among the acquired CT projections, resulting in blurring, ghosting, streaking artifacts, and distortion in reconstructed images. Motion artifacts are

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more prevalent in portable head CT which is often used in emergency room due to prolonged scanning time and critically ill patients who are unable to cooperate or remain still. Hardware solutions such as faster gantry rotations, dual-source systems, and motion tracking systems [2] can mitigate motion artifacts, but their integration into portable or C-arm CT scanners is still challenging. Therefore, there is a critical need to develop software solutions to correct head CT motion artifacts in clinical practice.

Existing software solutions can be categorized into two strategies. The first strategy compensates motion by estimating head motion during the acquisition using CT projection data. Solutions within this strategy include using 3D-2D registration to align each 2D projection to a 3D prior image[3], [4], employing iterative process to optimize image-based motion artifact metrics like total variation norm[5] or image entropy and sharpness[6], and utilizing partial angle reconstruction[7], [8], [9] techniques to estimate motion. However, obtaining and processing projection data in clinical settings remains challenging, which limits the practical application of these projection-domain-based methods. The second strategy relies solely on correction within the image domain. Despite the lack of data fidelity, image-domain-based solutions don't require projection data and are thus more practical for daily use. Current literature on head CT motion correction using such approaches is sparse, with existing solutions mainly utilizing convolutional neural networks (CNNs) trained on motion-free and motion-corrupted image pairs. Ko et al.[10] augmented U-Net with attention mechanism to selectively amplify or attenuate residual features, while Su et al.[11] integrated ResNet and dilation techniques into U-Net to retain high resolution features. However, these approaches have typically been validated on 2D motion (using CT slices as model input) or under conditions of slight motion. Empirically, we observe that these CNN-based methods tend to over-smooth images when addressing moderate to severe 3D motions.

Score-based diffusion models have recently excelled in various generative and inverse problems[12], [13], [14], [15], offering improved image quality and more stable training. This success is due to diffusion models learning advanced data representations by understanding the gradient of the log density of the datasets (i.e., data score). However, applying diffusion models to CT motion correction remains unexplored. Therefore, we propose the first diffusion model-based 3D head CT motion correction approach, named HeadMotion-EDM (HM-EDM). This method uses the motion-corrupted CT image as a conditional input and corrects the motion artifacts purely based on the image domain. The model successfully addresses two challenges in solving 3D motion artifacts using diffusion models. Firstly, correcting 3D motion requires a 3D diffusion model that takes 3D CT data as input, which can easily exceed GPU memory limits. To tackle this, we utilize a 3D-patch-wise diffusion training that maintains 3D information while reducing computational demands. Secondly, the standard denoising diffusion probabilistic model (DDPM)[12] may experience suboptimcal data score estimation by the trained neural network as well as the inaccurate approximation of true noise removal trajectory when using fewer steps in accelerated sampling[16], [17]. To overcome these issues, we leverage the advanced Elucidated Diffusion Model (EDM)[16] framework, which integrates several pivotal engineering improvements in both score-model training and the faster sampling process.

The contributions of our work are threefold. First, we propose HM-EDM to correct 3D head CT motion artifacts with efficient training and a faster sampling process. Second, we demonstrate that HM-EDM surpasses CNN-based techniques and standard DDPM in both simulation and phantom studies. Third, we apply HM-EDM to real-world portable head CT scans to show its practical application.

2 Methods

2.1 Score-based Diffusion Model for Motion Artifact Correction

We frame CT motion correction as a conditional generation task, sampling from a conditional distribution $p_{data}(x_0 \mid c)$, where $\{x_0, c\}$ is a pair of motion-free (x_0) and motion-corrupted (c, produced by filtered back-projection) CT images. Regarding the diffusion model, it has a forward and a sampling process. The forward process progressively adds Gaussian noise to the motion-free CT image, x_0, sampled from $p_{data}(x_0 \mid c)$, until making it indistinguishable from pure noise. This forward process could be characterized by a stochastic differential equation (SDE),

$$dx_t = f(t)x_t dt + g(t)dw_t \tag{1}$$

where $f(t)$ and $g(t)$ are the drift and diffusion coefficient, respectively, and w is a standard Wiener process. $t \in [0, T]$ denotes a time variable such that if defining the marginal probabilities of x_t as $p_t(x \mid c)$, then $p_0(x \mid c)$ equals to the original data distribution p_{data} and $p_T(x \mid c)$ equals to a Gaussian distribution. We set $T = 1000$ in this study. Specially, the transition probability from x_0 to x_t is given by:

$$p_t(x_t \mid x_0, c) = \mathcal{N}\left(x_t; \mu_t x_0, \mu_t^2 \sigma(t)^2 I\right) \tag{2}$$

with the noise schedule $\sigma(t) = t$ which defines the desired noise level at time t.

In the sampling process, the image generation starts from pure noise drawn from the prior distribution p_T and iteratively removes the noise. This process is expressed as the reverse SDE[13], [18]:

$$dx_t = \left[f(t)x_t dt - g(t)^2 \nabla_{x_t} \log p_t(x_t \mid c)\right]dt + g(t)d\overline{w}_t \tag{3}$$

Here, the gradient of the log density of the datasets, $\nabla_{x_t} \log p_t(x_t \mid c)$, is the score function of the dataset. Eq. 3 has a corresponding "probability flow" ordinary differential equation (ODE) [16] which eliminates randomness in the process and is given in a form generalized for any noise schedule $\sigma(t)$ as:

$$dx_t = \left[-\dot{\sigma}(t)\sigma(t)\nabla_{x_t}\log p_t(x_t \mid c)\right]dt \tag{4}$$

To solve Eq. 4, the score function $\nabla_{x_t}\log p_t(x_t \mid c)$ needs to be determined. Practically, we train a neural network to estimate the score function by score matching with a denoiser function $D(x_t; \sigma(t), c)$ so that:

$$\nabla_{x_t}\log p_t(x_t \mid c) = [D(x_t; \sigma(t), c) - x_0]/\sigma(t)^2 \tag{5}$$

This neural denoiser D isolates the noise component from the signal x_t and can be trained by minimizing the expected L_2 loss for samples drawn from p_{data} for every σ,

$$L := \mathrm{E}_{p_{data}}\mathrm{E}_{n \sim \mathcal{N}\left(0, \sigma(t)^2\right)}\left\| D(x_0 + n; \sigma(t), c) - x_0 \right\|_2^2 \tag{6}$$

n is the noise. Overall, in our proposed diffusion model HM-EDM, x_0 is the targeted motion-free image, and we train a neural denoiser $D(x_0 + n; \sigma(t), c)$, with the motion-corrupted image c as conditional input, to gradually remove noise n and eventually generates the motion-corrected image.

2.2 Patch-wise Model Training

Regarding the model architecture of neural denoiser D, we adapt the DDPM++ U-Net [13] for 3D data by changing original 2D convolutional layers and attention modules to 3D. However, we found that training such a 3D U-Net directly on 3D head CT data exceeds the GPU memory limitation. Further, this method requires gathering extensive 3D CT data. To reduce memory usage and the need for a large dataset, we leverage a patch-wise model training strategy:

$$L_{patch} = \sum_{i=1}^{N} \mathrm{E}_{p_{data}}\mathrm{E}_{n \sim \mathcal{N}\left(0, \sigma(t)^2\right)}\left\| D\left(x_{0,\phi_i} + n; \sigma(t), c_{\varnothing_i}\right) - x_{0,\phi_i} \right\|_2^2 \tag{7}$$

where x_{0,ϕ_i} denote a sub-volume sampled from the motion-free 3D CT data, x_0, which is the union of N such sub-volumes. The patch-wise loss, L_{patch}, accumulates the losses over these sub-volumes. While trained on patches with reduced height and width, the network can work on the original dimension during sampling due to the shift-invariance property of convolution layers. The implementation details can be found in section 2.4.

2.3 Advanced EDM framework

Our model uses the advanced EDM framework instead of the standard DDPM for two reasons. First, it is time-intensive for the DDPM sampling to go through the same number of timesteps (i.e., $500 \sim 1000$ steps) as in the forward process, while accelerated sampling using the reduced number of steps may lead to the truncation error and inaccurate approximation of true noise removal trajectory in DDPM. Second, the neural denoiser D in DDPM may not accurately estimate the data score. EDM[16] integrates several pivotal engineering improvements for these two challenges. Regarding the truncation error caused by a finite number of sampling steps, EDM uses Heun's 2nd order method as the numerical SDE solver, which introduces a correctional sub-step from x_t to x_{t-1} to account for change of dx/dt in the trajectory and thus align the sampling with the actual trajectory. Regarding the network training, a σ-dependent skip scaling is proposed to precondition the neural network, enabling the model to adaptively estimate either the noise n or the clean image x_0 or something in between based on the noise level σ. The rationale is that it is more effective to predict x_0 at higher noise levels and n at lower noise level[19], [20]. Therefore:

$$D(x_t; \sigma(t), c) = s_{skip}(\sigma(t))x_t + s_{out}(\sigma(t))F_\vartheta(s_{in}(\sigma(t))x_t; \sigma(t), c) \tag{8}$$

where F_ϑ is the network to be trained, s_{skip} denotes the skip scaling factor controlling model output type. When σ is large, s_{skip} is approaching to 0 so that F_ϑ directly predicts the denoised image, x_0; When σ is small, s_{skip} is approaching to 1 so that F_ϑ predicts the negative of noise, $-n$. s_{in}, s_{out} are other scaling factors to scale the input and output to unit variance [21]. By integrating the EDM framework, our model successfully reduces the sampling steps to 50—just 1/20th of the DDPM—s steps $T = 1000$ as in section 2.1)—while achieving a more accurate estimation of the data score function.

2.4 Implementation Details

For image preprocessing, all 3D CT data were resampled to a pixel dimension of [1, 1, 2.5]mm and reshaped into $256 \times 256 \times 50$ slices covering from the nose to the top of the head. During patch-wise training, patches sized $128 \times 128 \times 50$ were randomly extracted from the CT volume with necessary data augmentation and intensity range normalized to [−1,1]. For sampling process, the original x and y dimensions were preserved and non-overlapping stacks of 25 slices were processed sequentially to prevent exceeding the GPU's capacity. Sampling a complete 3D head CT requires two such stacks. Using patch-wise training, our HM-EDM method requires 32GB of memory for training with a batch size of 1 patch. The sampling needs 26GB to sample a $256 \times 256 \times 25$ stack. With faster sampling enabled by the EDM framework, we can sample a complete 3D head CT dataset in less than two minutes on a DGX-A100 GPU (NVIDIA, USA), whereas the conventional DDPM approach typically takes around 20 minutes on the same GPU.

3 Experiments

We trained our model using simulated data as we cannot acquire motion-static pairs from patients. We conducted a simulation study and a phantom study with known ground truth motion-free images to quantitatively compare the performance of HM-EDM with CNN-based methods and standard DDPM. Additionally, we conducted a reader study using a real-world portable head CT dataset, reviewed by two radiologists, to demonstrate the practical application of HM-EDM.

Simulation Study.

We conducted a simulation study with pairs of ground truth motion-free images and simulated motion-corrupted images. We retrosepctively collected 100 motion-free multi-slice head CT scans from a single medical center. For the simulation, we utilized a head motion model proposed by Jang et al.[6] and Chen et al.[9]. The head posture was described using 6 rigid motion variables including translations and rotations along the three image axes. Each motion variable was modeled using a B-spline with 5 control points equally spaced across one gantry rotation. The motion amplitude (i.e., the value of each control point) was randomly sampled from [−5, 5]mm or degrees for each translation or rotation variable, covering voluntary head motion commonly seen according to Kyme et al.[1]. The motion was applied to the motion-free image, and a simulated process of forward projection and filtered backprojection (FBP) was conduced using the actual CT geometry parameters from the scanner. We split the 100 scans into a training/validation (80 scans) and a testing (20 scans) dataset. Each scan was used to produce 50 simulations in training and 5 simulations in testing. The models were trained on the simulated training data.

We compared our HM-EDM method with the standard DDPM as well as the CNN-based methods (CNN with ResNet[11] and attention[10]). All methods use the same implementation details illustrated in section 2.4. The quantitative outcomes shown in Table 1 indicate that HM-EDM achieves the lowest MAE and the highest SSIM both in the brain region (MAE drops from 22.7±5.6HU in motion-corrupted images to 11.0±2.1HU, SSIM increases from 0.17±0.06 to 0.51±0.08) and overall image (MAE drops from 179±39HU to 52±13HU, SSIM increases from 0.64±0.10 to 0.95±0.03) compared to other competing methods ($p<0.001$ by one-tailed paired student t-test). Table 1 also shows that only HM-EDM preserves the contrast-to-noise ratio (CNR) in the brain region comparable to motion-free images (9.34±1.14 in motion-free vs. 9.47±1.06 in HM-EDM), whereas CNN-based methods tended to over-smooth the image, resulting in elevated CNR. Additonally, Fig. 2 showcases examples of motion correction in brain and skull, recovering fine structures in brain and skull anatomy.

Phantom Study.

An ACS head phantom (Kyoto Kagaku, Japan) was scanned using an CereTom Elite portable CT scanner (Neurologica, USA) with x-ray tube energy of 120 kVp and tube current of 20mA. To simulate motion, we stacked sinograms from multiple static acquisitions with varied head postures. These composite sinograms were reconstructed into 20 motion-corrupted images with different motion patterns. Without any fine-tuning, Table 2 shows that HM-EDM outperforms other competing methods in the brain correction, as well as preserves CNR in the brain region.

Real-world Application.

We retrospectively collected 5 portable head CT scans acquired by CereTom Elite portable CT scanner (Neurologica, USA) from the ICU patients. All scans were identified to have motion artifacts. We applied the trained HM-EDM directly without fine-tuning. The original and motion-corrected images were randomly assigned as image A and image B. Two radiologists, blinded to the image identities, compared A against B and scored the comparison regarding motion artifacts, diagnostic value, and preference in the range of −2 to +2 (where 0 means no difference, 1 means effective improvement and 2 means significant improvement; negative means degradation). Fig. 3 shows the HD-EDM results of all 5 scans. The motion corrected CTs always have positive scores when compared to the original CT on all scans for all 3 criteria.

4 Conclusion

This work introduces the first application of a diffusion-based generative model for CT motion artifact correction. Our proposed HM-EDM leverages a conditional diffusion model to significantly reduce motion artifacts in 3D head CT scans. Our method outperforms CNN-based methods and DDPM in both simulation and phantom studies. Furthermore, we conducted a pilot reader study using a small number of real-world clinical portable head CT scans to preliminarily demonstrate the practical use of our algorithm. However, this reader study lacks ground truth motion-free images. Therefore, future work will focus on larger real-world validation, where within one study, the patient has an initial motion-compromised scan as well as a subsequent motion-free rescan.

Prospect of application:

HM-EDM will be practically useful as an image postprocessing tool to correct motion artifacts in head CT scans. It is accurate, fast (<2 minutes/case), and doesn't require the use of CT raw data. It will be particularly useful for portable head CT scans in emergency rooms, which more commonly have motion artifacts. Our reader study has preliminarily confirmed its effectiveness.

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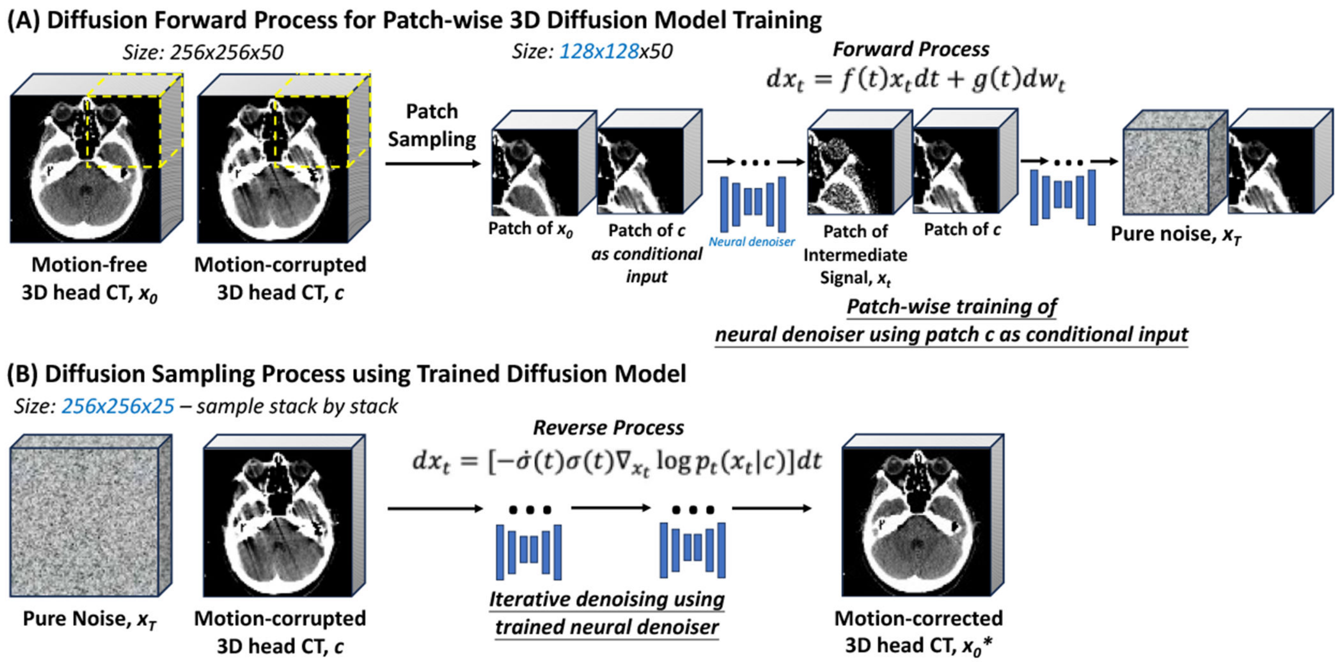

Fig. 1.

The overall pipeline of HM-EDM. (A) The forward process involves patch-wise training of the neural denoiser, which takes conditional input in the form of patches consisting of the intermediate signal x_t along with the motion-corrupted image c. (B) the sampling process begins with the pure 3D noise x_T and iteratively removes noise using the trained denoiser to generate motion-corrected 3D head CT image.

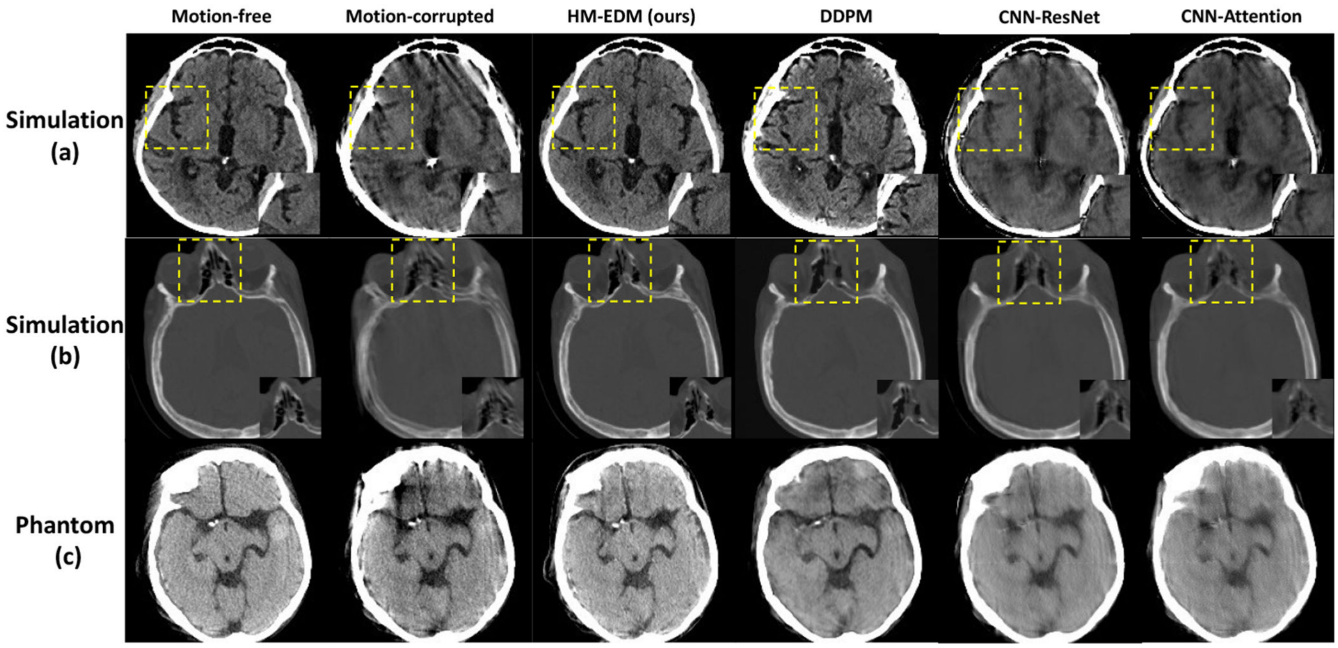

Fig. 2.
Performance on simulation study (a,b) and phantom study (c). (a) illustrates correction within the brain tissue, showcasing our method's ability to restore the visibility of fine structures as indicated by the yellow box. (b) shows correction in a severe skull ghosting artifact instance. (c) shows the results on the head phantom.

	(a)		(b)		(c)		(d)		(e)	
	R1	R2	R1	R2	R1	R2	R1	R2	R1	R2
Motion artifact	1	2	2	2	2	2	1	1	1	1
Diagnostic value	2	2	2	1	2	2	1	1	1	1
Preference	2	2	2	2	2	2	1	2	1	1

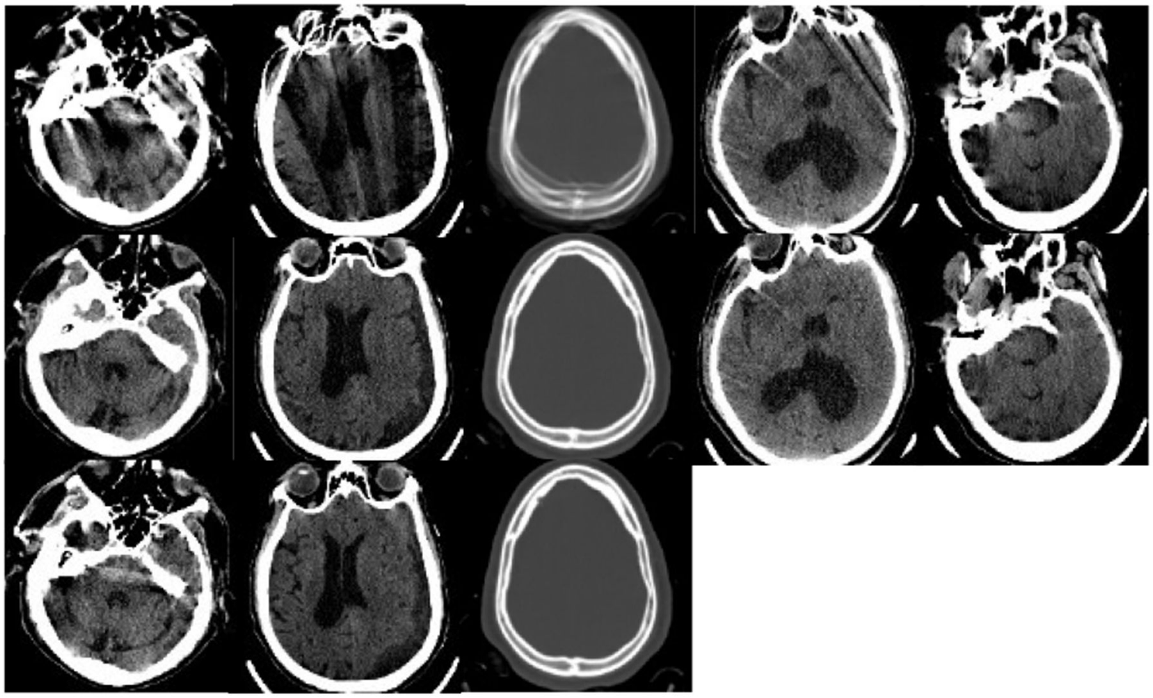

Fig. 3.

Performance on 5 real-world portable CT scans. (a) and (b) show the correction of severe artifacts in brain tissue where (b) also recovers the chronic hemorrhage, while (c) shows the correction of severe “double-skull” artifacts. For these three severe cases, by comparing the adjacent slice in the original image (with much less motion artifact) in the third row, we can see that our HM-EDM model potentially utilizes information from these adjacent slices for 3D anatomical context. (d) and (e) show two examples with slight/moderate streaking artifact (d) and star-like artifact originating from the skull (e). At the top, we show scores by two radiologists comparing the corrected image to the original image. Scores range from +1 (effective improvement) to +2 (significant improvement).

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Table 1.

Simulation study's quantitative results. Metrics for the brain are measured on pixels within [0,100]HU in the motion-free image, and metrics for overall are measured on foreground pixels >−100HU. MAE are measured in Hounsfield Units (HU). CNR = contrast-to-noise ratio. The CNR in the brain region was measured in a ROI of 50×50 pixels from brain center, approximating the contrast between white matter and ventricle. A higher CNR indicates a smoother image. **The CNR for motion-free images is 9.34±1.14**.

ROI	Metric	Motion-corrupted	HM-EDM (ours)	DDPM[12]	CNN-ResNet[11]	CNN-attention[10]
Brain	MAE↓	22.7±5.6	**11.0±2.1**	12.6±2.8	16.6±4.3	17.0±3.6
	SSIM↑	0.17±0.06	**0.51±0.08**	0.41±0.09	0.25±0.06	0.23±0.05
	CNR↓	8.44±1.54	**9.47±1.06**	9.84±1.31	10.69±2.09	10.99±2.14
Overall	MAE↓	179±39	**52±13**	69±19	74±18	76±19
	SSIM↑	0.64±0.10	**0.95±0.03**	0.92±0.04	0.92±0.03	0.91±0.04

Table 2.

Phantom study's quantitative results. The CNR for the motion-free image is 8.64.

ROI	Metric	Motion-corrupted	HM-EDM (ours)	DDPM[12]	CNN-ResNet[11]	CNN-attention[10]
Brain	MAE↓	21.5±2.2	**12.3±0.9**	13.6±1.3	18.4±1.9	18.6±2.0
	SSIM↑	0.38±0.03	**0.57±0.06**	0.57±0.06	0.40±0.03	0.41±0.03
	CNR↓	8.47±0.07	**8.92±0.18**	9.94±0.33	11.63±1.26	12.33±1.66
Overall	MAE↓	116±17	**84±10**	85±8	101±12	99±13
	SSIM↑	0.83±0.04	**0.86±0.02**	0.86±0.02	0.84±0.03	0.85±0.03